Downsizing of a Pulsatile Total Artificial Heart—The Effect on Hemolysis

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A downsized version of the ReinHeart total artificial heart (TAH) was developed. Hemocompatibility needs to be revised since the operating point of the downsized TAH has changed to a higher pump frequency to accomplish the same cardiac output. A mock circulation loop was designed, containing a left side for hemocompatibility testing and a right side to mimic realistic work conditions. A protocol for hemolysis testing was established using pooled porcine blood with an operation point of 5 L/min, a mean outlet pressure of 100 mm Hg and a mean inlet pressure of 12 mm Hg. Six trials were performed testing two downsized TAH (one with a compliance chamber [CC] connected, necessary for a pneumatic decoupling of both membranes and one open to atmosphere) and a BPX-80 as reference pump. The average modified index of hemolysis and normalized index of hemolysis (NIH in mg/100L) from six individual trials of the reference pump were 0.34 (0.07) and 3.21 (0.61) and of the TAH open to atmosphere 4.18 (1.19) and 38.85 (10.59), respectively. In between TAH with and without CC, there was no significant difference. A NIH ratio of TAH and reference pump was calculated to minimize variation of the different blood batches used in individual trials. Due to the downsizing, the ReinHeart's hemolysis level increased by around 22% compared with the original size version. Comparing the results to clinically approved left ventricular assist devices, the level of hemolysis can still be considered acceptable. *ASAIO Journal* 2022; 68;34–40

According to the World Health Organization, cardiovascular diseases are the number one cause of death globally in 2017. Due to the chronic lack of donor hearts, options of alternative therapies such as mechanical circulatory support are rising. Especially in patients with biventricular end-stage heart failure, total artificial heart (TAH) therapies are becoming more and more important [1]. Currently, the pneumatic-driven, partially implantable SynCardia 70cc (SynCardia Systems Inc., Tucson, AZ) is the only TAH system approved by the US Food and Drug Administration. New fully implantable devices are currently under development[2] to further improve patient's quality of life.

In recent years, our department developed a new ReinHeart TAH design in cooperation with the ReinHeart TAH GmbH, facing the major challenge of the limited space inside the human body. Described by Pelletier *et al.*,[3] the ReinHeart TAH uses the principle of an electromagnetically driven piston displacement pump. First studies demonstrated its proof of principle.[3–8] The new downsized ReinHeart TAH uses the same principle as already described in literature but with its implantable pump unit reduced in volume by 20% to around 390 ml and a new stroke volume of around 34 ml (compared with 41 ml of the original design). A comparison of both designs can be seen in Figure 1. Consequently, the pump operates at higher pump rates to generate the same cardiac output (CO) as its original version. Hemocompatibility needs to be revised, considering this new operation point. Furthermore, when fully implanted, a compliance chamber (CC) is needed[6] whose influence on hemolysis has not been tested yet. It could affect the valve closing speed and therefore induce more hemolysis.[9]

In our study, the effect on hemolysis of a downsized pulsatile displacement TAH was investigated in several trials. Additionally, the influence on hemolysis of a CC prototype connected to the TAH was analyzed.

This paper presents the experimental method used to investigate the effect of design change of the TAH on red blood cell damage. Orientating on the standard practice for assessment of hemolysis in continuous flow blood pumps ASTM F 1841 and a presented method by Graef *et al.*[5] for investigation of hemolysis in a pulsatile TAH, a specific mock circulation loop has been designed and applied. To address considerable differences between individual blood batches, a ratio of hemolysis between the tested TAH and a reference pump was calculated for every individual trial.

Material and Methods

Description of the New ReinHeart TAH 2.0

The ReinHeart TAH system uses the principle of a passive filling piston displacement pump to replicate the function of a natural heart. Its central element is the pump unit.[10] As described by Pelletier *et al.*,[3] it consists of two artificial ventricles and a linear actuator as the drive unit in between. The space available for an implantation of such a device is limited.[11] To fit a wide range of patients, a smaller device was developed.

The new downsized ReinHeart TAH pump unit uses the same principle as the version before but is reduced in volume by 20%. To obtain the same CO, the TAH pump rate had to be increased from 120 bpm to 145 bpm. The new ventricles are made from the same materials as described in literature.[5] The described mechanical heart valves were changed to CryoLife

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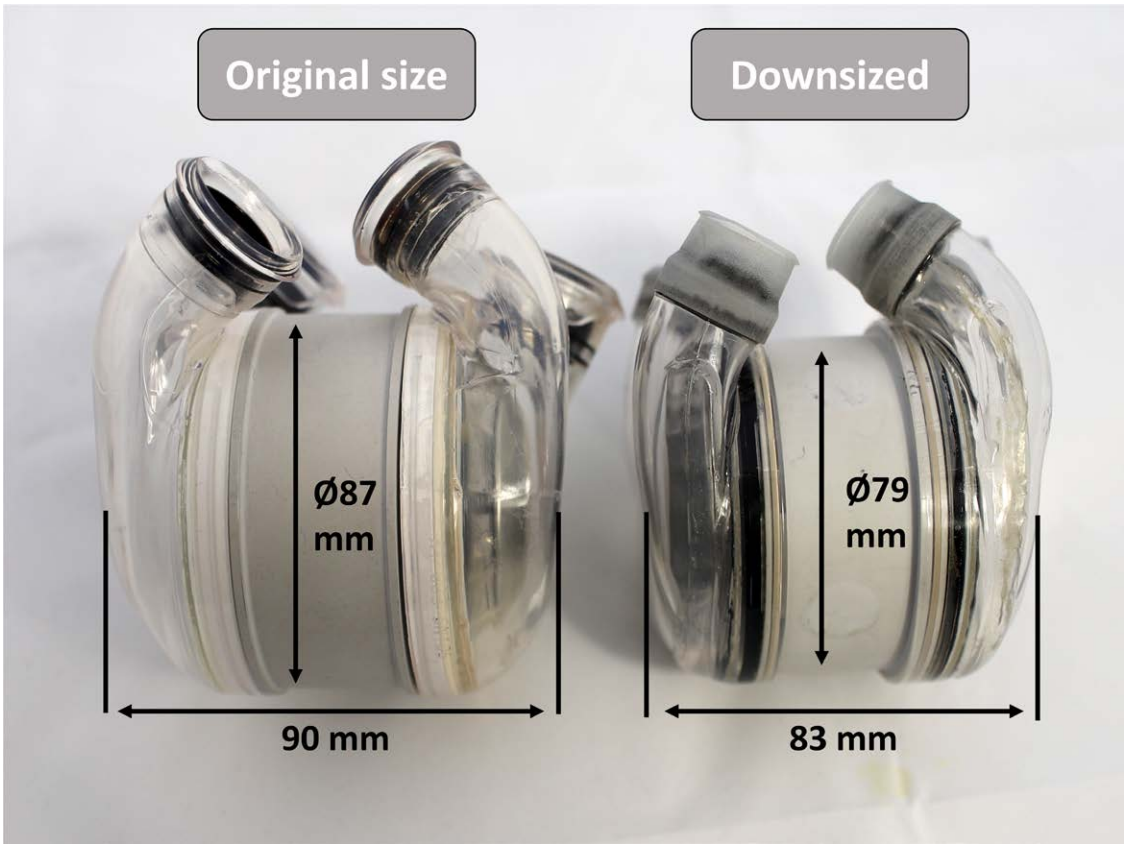


Figure 1. Comparison of the original size TAH (left) and the downsized TAH (right). TAH, total artificial heart. full color online

On-X (CryoLife Inc., Kennesaw, GA) bileaflet mechanical valves using an ONXA-23 valve for left inlet (mitral position) and ONXA-19 valves for the other three positions.

An additional CC connected to the TAH is still mandatory when implanted to achieve a pneumatic decoupling of both membranes and thereby ensure a passive filling of each pump chamber during its diastole.[6] A hermetically sealed drive unit could provoke suction of the atria, higher inflow acceleration and faster closing of the mechanical heart valves. A prototype of a fully implantable CC was designed and its influence on hemolysis was investigated in comparative trials. In the following sections, TAH with such a device connected are called downsized TAH CC. All other downsized TAH were tested with their drive unit open to atmosphere representing an ideal CC.

Test Loop Design

Graef *et al*.[5] proposed a mock circulation loop to test hemocompatibility of a separated ventricle of a pulsatile TAH. The same principle is used in this study, seen in Figure 2. To ensure the correct behavior of the tested full TAH system, the mock circulation loop consists of a left- and right-side loop, separated from each other and connected to the corresponding TAH ventricles. Both circulation loops consist of the same major elements such as an arterial CC, a flow resistor and an atrium. Only the left-side loop was filled with blood and used for hemolysis testing.

Based on recommendations by Graef *et al*.,[5] the pressure was measured inside the CC. It was possible to reduce the filling volume of the loop and the distance between outlet of the TAH and arterial CC to minimize the influence of the fluid's inertia, as seen in Figure 3. All chosen blood-facing materials were hemocompatible. The left-side loop consists of two nondilated silicon tubes made of Elastosil RT 625 (Wacker Chemie Ag, Munich, Germany) as well as four PC fittings and 70 cm of medical-grade 3/8″ PVC tubing (Raumedic AG, Helmbrechts, Germany). A heat exchanger (Eco Silver, Lauda Dr. R. Wobser GmbH, Lauda-Koenigshofen, Germany) was connected to the outer part of the left atrial chamber and controlled manually to keep the blood temperature at 37°C ± 1°C. The blood was separated from the heat exchanger's fluid by one of the nondialated silicone tubes. The maximum temperature of the heat exchangers never exceeded 39°C to avoid heat-related blood damage. Clinical-grade pressure sensors (DPT-6000; Codan pvb Critical Care GmbH, Lensahn, Germany) were used to monitor the pumps inlet and outlet pressure for both sides during the entire test period. The flow was measured using ME9PXL16 ultrasonic flow probes in combination with T402 Multi Channel Research Consoles (Transonic Europe B.V., Elsloo, Netherlands).

Blood Preparation

To achieve the best comparability of this study to the published study of the original size TAH by Graef *et al*.,[5] the blood preparation was performed in the same way. For each trial, 2 L of freshly collected (<2 h before start) porcine blood were used. The blood was collected during the slaughtering process in a slaughterhouse from two pigs. For each liter of

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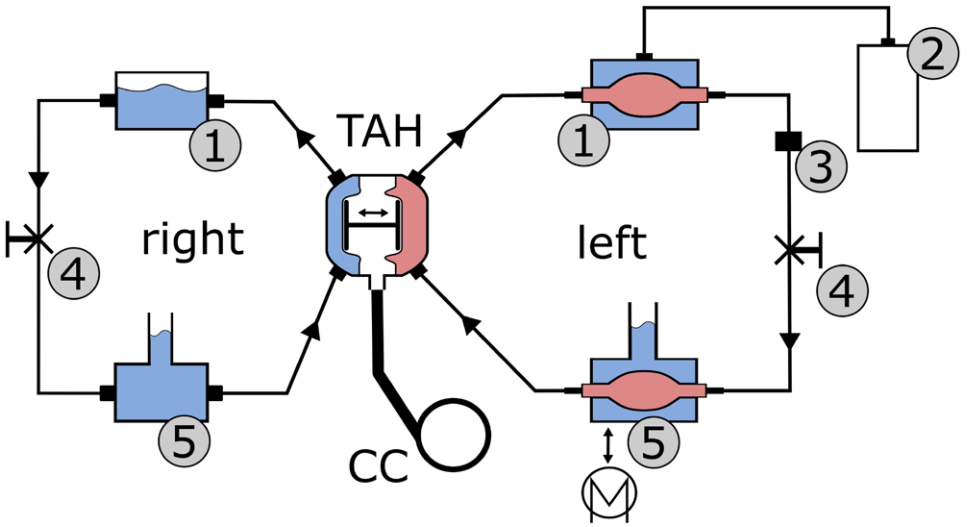


Figure 2. Mock circulation loop (right loop filled with water, left loop filled with blood): total artificial heart in the center with a connected compliance chamber; 1) arterial compliance chamber; 2) air compartment; 3) flow sensor; 4) flow resistor; 5) atrium and heat exchanger. TAH, total artificial heart. full color online

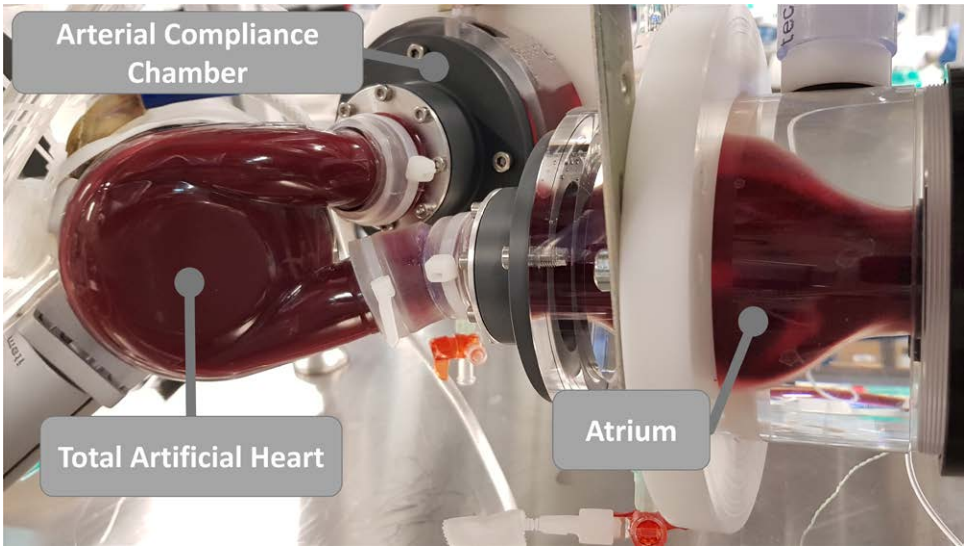


Figure 3. Left ventricle of the TAH connected to the left side of the mock circulation loop and filled with blood. TAH, total artificial heart. full color online

blood, 15,000 international units of sodium heparin (B. Braun, Germany), 1.8 ml glucose (50%), 1.6 ml antibiotics, and 50 ml saline solution (0.9%) were added. The blood of both animals was pooled in a blood bag afterward. By adding more saline solution (0.9%), the hemoglobin concentration (Hb) was set to 9.5 ± 1 g/dl with a corresponding hematocrit (Ht) of 30 ± 2%. The base excess was set to 0 ± 4 mmol/l with the addition of bicarbonate. For each test loop, the blood was infused by gravitational force through a sampling port of the blood bag as proposed in the standard practice for assessment of hemolysis in continuous flow blood pumps ASTM F 1841.

Test Protocol

Taking the large expenditure of such a trial into account and the recommended number of at least five tests by the ASTM F 1841, six individual trials were performed. Besides the static control, with each trial, a downsized TAH, a downsized TAH CC, and a reference pump were investigated simultaneously in three identical test loops. As a reference pump, a BPX-80 Bio-Pump (Medtronic, Minneapolis, MN, USA) was used connected to the left-side loop. According to ASTM F 1841, each loop was primed with saline solution for several minutes. After its removal, the loops were filled with 480 ml of blood immediately and all air was eliminated. By gradually increasing pumping speed and adjusting the resistor, the operation point of the left-side loop with a mean flow of 5 ± 0.25 L/min and a mean outlet pressure of 100 ± 5 mm Hg were reached. For the TAH loops, the outlet pressure of the right-side loops was set to 25 ± 5 mm Hg accordingly. To assure a complete filling of the TAH ventricles, the inlet pressure of both sides was set to a mean pressure of 12 ± 3 mm Hg. The same values were set for the reference pump. The hydraulic test parameters were kept within its previously

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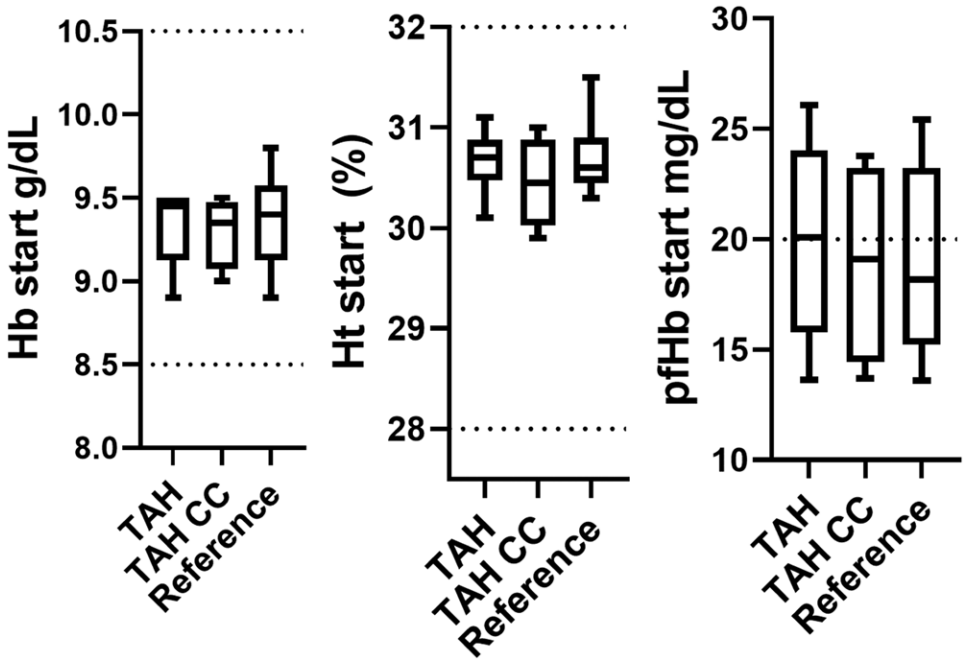


Figure 4. Test loop parameters of the downsized TAH with and without a connected CC and the reference pump BPX-80. A mean value of every parameter was taken at every sampling point for all trials. The values are displayed as a box plot with whiskers going from min to max. The dotted lines indicate the defined thresholds. CC, compliance chamber; TAH, total artificial heart.

mentioned thresholds during the entire test period by only adapting the flow resistor if necessary. All tests were performed for 6 hours while taking blood samples and mean values of the test loop parameters hourly. At every single sample point, 1 ml of blood was drawn and discarded before 3 ml of blood were drawn for analysis. The plasma free hemoglobin (pfHb) was determined with an Ultrospect 2100 pro photometer (GE Healthcare Europe GmbH, Freiburg Germany). An XT2000i Vet (Sysmex Sverige, Kungsbacka, Sweden) and an ABL825 Flex (Radiometer GmbH, Krefeld, Germany) were used for a differential blood count and a full blood analysis. After each experiment, the tubing and silicone parts of the loop were discarded. The TAH ventricles and the pump head of the BPX-80 were rinsed with water and cleaned in pepsin/citrate solution before reuse. The normalized index of hemolysis (NIH) was calculated by

$$\mathrm{NIH}\left(\mathrm{mg}/100\ \mathrm{L}\right) = \Delta\mathrm{freeHb} * V * \frac{100 - \mathrm{Ht}}{100} * \frac{100}{Q * T}$$

with ΔfreeHb as the increase of plasma free hemoglobin (mg/L) over the sampling time interval, V the circuit volume (L), Q the flow rate (L/min), Ht the hematocrit (%), and T the sampling time interval (min). In addition, the modified index of hemolysis (MIH) was calculated by

$$\mathrm{MIH} = \Delta\mathrm{freeHb} * V * \frac{100 - \mathrm{Ht}}{100} * \frac{10^6}{Q * T * \mathrm{Hb}}$$

wherein Hb is the total blood hemoglobin concentration at time zero (mg/L).

For each trial, NIH and MIH were calculated for every equidistant sample point.[1–6] The presented NIH and MIH of every individual trial were calculated by taking the average of these six values. All other mean results presented are followed by their standard deviation in parenthesis.

Results

Mock Circulation Loop

Using the described mock circulation loop, it was possible to keep the important test loop parameters mainly within their thresholds. A mean value of the last 10 seconds of every measured parameter was taken at each sample point for every pump loop throughout all six experiments and plotted in Figure 4. The dotted lines indicate the defined thresholds. Overall the mean flow of all test loops was 5.0 (0.2) l/min with a mean outlet pressure of 99.7 (2.1) mm Hg, a mean inlet pressure of 12.1 (2.1) mm Hg and a mean blood temperature of 36.9°C (0.3). Every loop reached the desired 37°C ± 1°C within the first half-hour. The blood temperature at filling was 35.9°C (1.5) for all loops with 33°C at lowest.

Figure 5 shows the mean blood values for every pump at time zero with an overall mean hematocrit of 30.63% (0.38), a mean hemoglobin 9.34 (0.24) g/dl and a mean plasma free hemoglobin of 19.32 (4.31) mg/dl.

Hemolysis

Figure 6 shows the plasma-free hemoglobin for all trials over the course of time as proposed by the ASTM F 1841. The static control does not show any unusual changes of plasma-free hemoglobin in any trial. During the six hours, all pump curves show a mostly gradually constant increase. There was no thrombus formation detected in any part of the loop or pump at any time due to the use of heparinized blood. Furthermore, no significant change in hematocrit or number of platelets was noticed.

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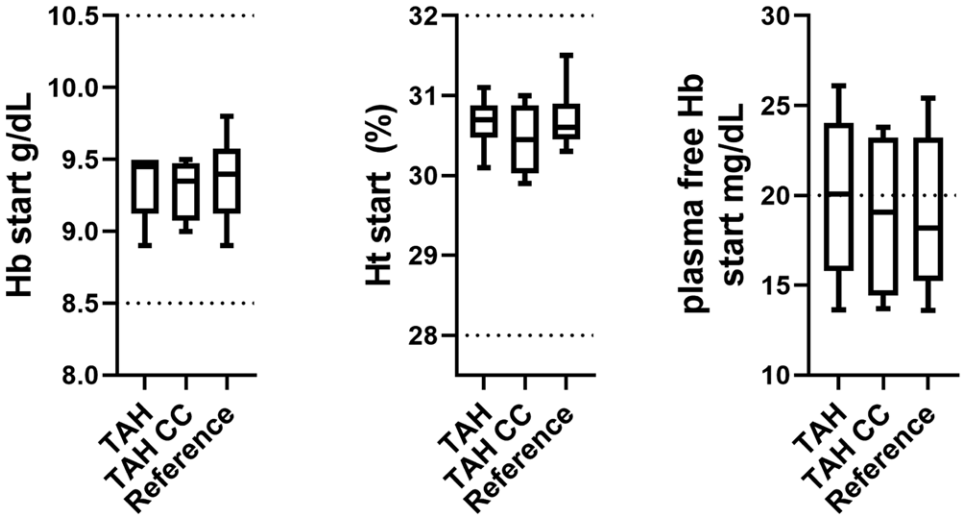


Figure 5. Box plot of blood parameters at start for the downsized TAH with and without a connected CC and the reference pump BPX-80. The dotted lines indicate the defined thresholds. CC, compliance chamber; TAH, total artificial heart.

The calculated NIH of every tested device for each trial is demonstrated in Figure 7. The mean NIH of the six downsized ReinHeart TAH open to atmosphere is 38.85 (10.59) and the mean NIH of the BPX-80 reference pump is 3.21 (0.61). Due to CC-related complications during the first two trials, there are no results for the downsized TAH CC. Considering only trial 3–6, there are no significant differences between the TAHs open to atmosphere (mean NIH 44.17 [8.58]) and the ones with a passive CC connected (mean NIH 35.72 [12.39]).

In Table 1, the NIH ratio of the tested downsized ReinHeart TAH open to atmosphere and the reference pump BPX-80 are presented for each trial and compared with data already published of the original size version of the Reinheart TAH. Overall the downsized TAH (12.01 [1.48], n = 6) has an 22.2% increased ratio compared with the original size TAH results (9.35, n = 1) presented by Graef *et al.*[5] Comparing only the mean NIH, the downsized TAH shows an increase of 30% compared with the original size version although the difference is not significant due to the relatively high variation. Furthermore, the MIH for every downsized TAH is listed (mean MIH 4.18 [1.19], n = 6) and the MIH TAH/BPX-80 ratio is calculated.

Discussion

Among other features, the downsized ReinHeart TAH design aimed to reduce the overall volume of the device by 20% to be available for more patients with limited mediastinal space. We are currently performing a human in silico and postmortem anatomic fitting study to validate the effect of the downsizing. However, smaller design comes with new challenges. Higher pump rates of 145 BPM instead of 120 BPM are necessary to obtain the same CO of 5 L/min under the described conditions. This leads to a more frequent opening and closing of the mechanical heart valves. Furthermore, higher shear stress might occur in the smaller inflow and outflow pathways and faster acceleration of TAH's membrane.

By using the presented specific mock circulation loop and test protocol, it was possible to investigate hemocompatibility

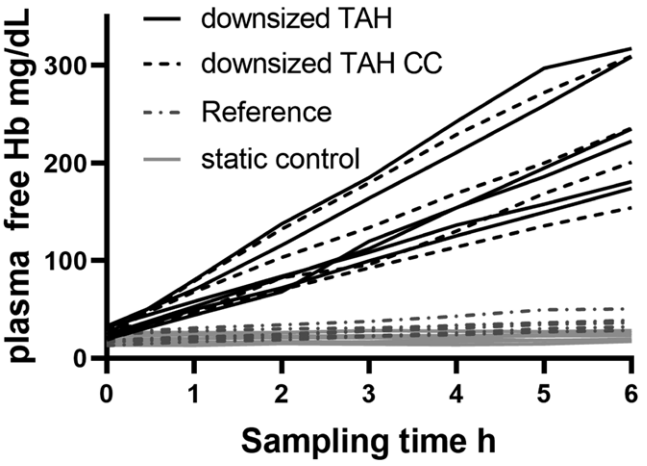


Figure 6. Plasma-free hemoglobin of all tested pumps including each static control over the sampling time. TAH, total artificial heart.

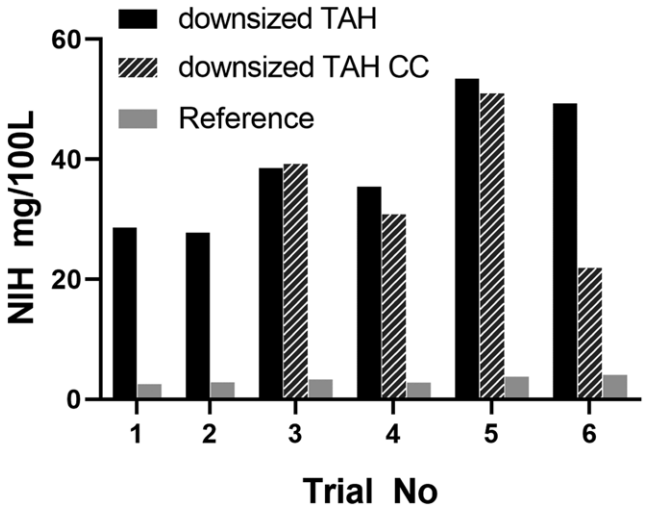


Figure 7. NIH of the devices for every trial. CC, compliance chamber; NIH, normalized index of hemolysis; TAH, total artificial heart.

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Table 1. Mean NIH and MIH Ratio (TAH/BPX-80) of the Downsized TAH and the Original Size TAH, Both Open to Atmosphere

	NIH TAH	NIH BPX-80	MIH TAH	MIH BPX-80	NIH Ratio	MIH Ratio
Graef *et al.*[5]	30* (n = 6)	–	–	–	–	–
	24.30*	2.60	–	–	9.35*	–
Trial 1	28.65	2.51	3.11	0.27	11.43	11.43
Trial 2	27.79	2.85	2.97	0.30	9.76	9.76
Trial 3	38.50	3.27	4.07	0.35	11.77	11.65
Trial 4	35.43	2.77	3.74	0.29	12.77	12.72
Trial 5	53.41	3.75	5.96	0.42	14.24	14.25
Trial 6	49.33	4.08	5.22	0.42	12.08	12.34

NIH is given in mg/100L.
*Original size TAH.
MIH, modified index of hemolysis; NIH, normalized index of hemolysis; TAH, total artificial heart.

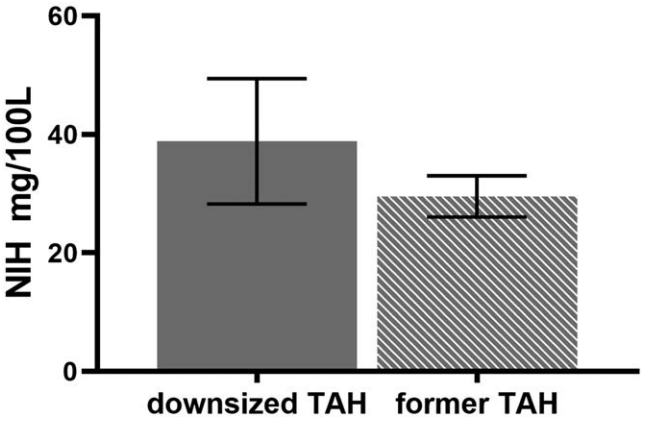


Figure 8. Comparison of the NIH of the downsized TAH and the original size TAH published in Graef *et al.*[5] NIH, normalized index of hemolysis; TAH, total artificial heart.

of the downsized ReinHeart TAH design showing an increase of overall mean NIH compared with the prior design. Taking all values of both studies into account, the difference in NIH between the two designs is not significant, although a plausible tendency can be seen as demonstrated in Figure 8. More hemolysis trials could increase the significance, but time, cost, and effort would be disproportionate. This study compares the hemolysis rate at a fixed flow rate of 5 L/min. It is known that the hemolysis rate correlates with the flow rate of the pump in rotary blood pumps.[12] Volume displacement pumps like the ReinHeart TAH might not be as sensitive to a change of flow rates in this regard. Nevertheless, an investigation of the hemolysis at other flow rates would be of interest for the further characterization of the device and will be the subject of future studies.

Consistent with our findings on the NIH of the BPX-80 used as reference pump, Kosaka *et al.*,[13] Nosé,[14] and Abe *et al.*[15] reported similar values. Taking the potentially high variance of blood from different donors in every trial into account, a NIH and a MIH ratio between the tested TAH and the reference pump were calculated for each trial. With this approach, the relative standard deviation between the individual trials was halved compared with the generally used approach of calculating an average NIH or MIH value of all individual trials. Comparing the NIH ratio of the new design to reported values of the original design by Graef *et al.*,[5] shown in Table 1, our findings of increased hemolysis level in the new design are supported with an increase of 22%. As a limitation of this comparison, it needs to be mentioned that Graef *et al.*, only performed a single trial with a BPX-80 as a reference pump; therefore, only a single NIH ratio was calculated.

Due to the use of fully heparinized blood in the hemolysis trials, a statement regarding the thrombogenicity of the ReinHeart TAH is yet not possible and needs to be investigated in a separate *in vitro* study.

Another aspect investigated in this study was the impact of a passive CC connected to the TAH, which will be necessary when fully implanting the TAH system. The results show no significant influence in the change of hemolysis between devices open to atmosphere and devices connected with a CC.

It could be considered as a limitation of our study that against recommendations of ASTM F1830 slaughterhouse blood was used. Furthermore, the plasma-free hemoglobin concentration at the start was not always below 20 mg/dl as recommended by the ASTM F1841. Nevertheless, we cannot see any significant influence of a higher pfHb start value on the results of our experiments. Similar observations have been reported in literature (Mueller *et al.*[16] and Graef *et al.*[5]). One reason for the relatively high pfHb start value could be the pooling of the blood. This was needed to have enough blood to fill all three loops with the same exact blood for best comparison. This approach is a standard method in our lab. We do not see any significant influence from the pooling. The same blood preparation was done by Graef *et al.*,[5] which leads to a better comparability of both studies (original size TAH *vs.* downsized TAH). Apart from this, only the hemocompatibility of the left ventricle was tested.

As another limitation, it could be mentioned that the outlet pressure of the right-side loops of the TAH is slightly high. A mean pressure of 25 ± 5 mm Hg in the pulmonary artery can be considered as pulmonary hypertension. With the used setup, it was not possible to decrease the outlet pressure any further without decreasing the inlet pressure of the right side as well. We think that it was more important to ensure a complete filling of the right ventricle (which otherwise could affect the movement of the left side) than having a lower outlet pressure (which mostly influences the overall power consumption of the pump in our study). Overall the raised pressure levels can be seen as a worst-case estimation compared with the possible later clinical situation.

Comparing the downsized ReinHeart TAH left ventricle mean MIH of 4.18 as shown in Table 1 to clinically approved left ventricular assist devices Heartmate 2 and Heartmate 3 (Abbott, Inc, Chicago, IL), the ReinHeart is in the same range. Bourque *et al.*[17] presented mean MIH values of 9.8 and 3.5 for Heartmate 2 and Heartmate 3 under comparable test conditions but using bovine blood. Even though experimental studies show similar responses to shear-induced hemolysis between porcine, bovine, and even human blood,[18] it needs to be mentioned that a direct comparison between those studies has its limitations.

In conclusion, the smaller design of the downsized ReinHeart TAH will help to fit a wider range of patients with limited mediastinal space. By using an enhanced test method for hemolysis testing in a pulsatile TAH, initially proposed by Graef *et al.*, it was possible to investigate the impact of the

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smaller pump design on hemolysis. Considering the big effort of such *in vitro* trials, the relatively small sample size might be acceptable. As described, the need of a higher pump frequency to obtain the same CO and the smaller diameter of in- and outflow pathways is accompanied with an increase of red blood cell damage. Downsizing the ReinHeart TAH 20% in volume shows an increased hemolysis tendency confirming the initial hypothesis. Comparing these results to clinically approved LVADs, the level of hemolysis can still be considered acceptable.

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